# Group Representation Theory Notes

## Giovanna Carnovale - SGSS
## Andriy Regeta - SGSS

Filippo Gargiulo*

Ottobre - Dicembre 2026

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*Mail per segnalare errori: filippo.gargiulo@studenti.unipd.it

# §1 Basics of Representation theory

**Definizione 1.1** (Group representation)**.** A *representation* of a group $G$ is a morphism $\rho : G \to \mathrm{GL}(V)$, where $V$ is a certain vector space. Commonly one refers to $V$ as being the representation of $G$, taking as granted from the context the mosphism $\rho$.

Analogously a representation of a ring or a field is a morphism from it to an arbitrary $\mathrm{GL}(V)$. As it's usual with actions, if $\rho$ is clear from the context we write

$$\rho_V(g)v = g \circ v = gv$$

with a slight abuse o notation.

**Definizione 1.2** (Group algebra)**.** Given a field $k$ and a group $G$ the *group algebra* of $G$ is the set

$$kG := \left\{ \sum_{g \in G} \lambda_g g : \lambda_g \in k \right\}$$

equipped with the obvious sum (as a vector space) and the distributive multiplication that satisfies the multiplication of $G$.

We should prove that the one above is actually an algebra, but it is quite easy indeed. Given a representation $\rho$ of $G$ we can extend it linearly to a representation $\tilde{\rho}$ such that

$$\tilde{\rho} : kG \to \mathrm{End}(V)$$

and this is a ring representation (we go in End because multiplication in $kG$ is not necessarily invertible).

**Definizione 1.3** (Algebra)**.** Given a field $k$, an *algebra* over it is a $k$-vector space equipped with a bilinear, unitary and associative product.

Just as with groups we define a representation of an algebra as a morphism $\rho : A \to \mathrm{End}(V)$.

**Esempio 1.4** (Trivial representation)

For any group the map $\rho : G \to \mathrm{GL}(V)$ such that

$$g \mapsto 1$$

is the so called *trivial representation* of $G$.

**Esempio 1.5** (Regular representation)

For any group $G$, if we take $V = kG$ we get the so called *regular representation* of $G$ as

$$h \circ \sum_{g \in G} \lambda_g g \mapsto \sum_{g \in G} h\lambda_g g.$$

**Esempio 1.6** (Representation of a permutation)
For $G = S_n$ and $V = k^n$ the map that permutes the canonical basis is a representation, in other words

$$g \circ e_i := e_{g(i)}$$

is a representation.

**Esempio 1.7** (Determinant)
Given a representation $\rho : G \to \mathrm{GL}(V)$ the map

$$\det \circ \rho$$

is also a representation (in particular is always 1-dimensional).

## §1.1 Morphisms of representations

**Definizione 1.8** (Morphisms of representations)**.** Given a group $G$ and two representations $\rho_V$ and $\rho_W$ over the same field $k$, a *morphism* between theese is a linear map $T : V \to W$ such that is compatible with both $\rho_V$ and $\rho_W$; that is, we have the following commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ {\scriptstyle \rho_V(g)}\downarrow & & \downarrow{\scriptstyle \rho_W(g)} \\ V & \xrightarrow[T]{} & W \end{array}$$

for any $g \in G$. Theese are also called *intertwined operators* or *G-equivariant maps*, or *G-morphisms*... they have a lot of names. Anyway we say that a morphism of representations is an *isomorphism* if it is invertible.

What we are saying essentially is, being that $\rho_V(g)$ and $\rho_W(g)$ are matrices, $T$ commutes with all such matrices in the way given above. The set of all intertwined operators is oftend denoted as

$$\mathrm{Hom}(V, W) \geq \mathrm{Hom}_G(V, W)$$

which is always a non empty vector space (!).
Let's now consider two representations $V$ and $W$, we can investigate what happens to the action of $G$ when we take $V \oplus W$, as we expected

$$g \circ (v, w) = (g \circ v, g \circ w)$$

and no one is surprised. As usual with vector spaces, given two finite dimensional representations $V, W$, if we choose basis $\mathcal{B}_V$, $\mathcal{B}_W$ theese induce a morphism of representations from $V$ and $W$ to

$$\rho_V : G \to \mathrm{GL}_n(k)$$

and

$$\rho_W : G \to \mathrm{GL}_m(k)$$

in this way $\rho_{V \oplus W}(g)$ has also a nice matrix representation

$$\rho_{V \oplus W}(g) = \left(\begin{array}{c|c} \rho_V(g) & 0 \\ \hline 0 & \rho_W(g) \end{array}\right).$$

**Lemma 1.9**

Given a map between two representations $V$ and $W$ of $G$, we have that this is $G$-equivariant iff is $kG$ equivariant, given $k$ the field over which $V$ is defined.

**Esempio 1.10** (Dual representation)

Given a representation $V$ of $G$, it's dual space $V^*$ has naturally a representation structure

$$(\rho_{V^*}(g)f)(v) = f\left(\rho_V(g)^{-1}v\right)$$

where the inverse is taken to make sure that it is an homomorphism.

*contenuto...*

a more compact way to write it is $g \circ f = f \circ g^{-1}$. An even more compact way to see is through matrices, choosing a basis $B$ and it's dual basis $B^*$ of $V^*$ we see that the actions, as matrices satisfy

$$\rho_{V^*}(g) = \rho_V(g)^\intercal$$

**Definizione 1.11** (Tensor product of representations)**.** Given two representations $V$ and $W$ of a group their *tensor product* is the space

$$V \otimes W := \langle \{v \otimes w : v \in V, w \in W\} \rangle$$

equipped with the action

$$\rho_{V\otimes W}(g)(v \otimes w) := \rho_V(g)v \otimes \rho_W(g)w.$$

**Osservazione 1.12.** Given spaces $W$ and $V^*$ there is always a canonical injection

$$F : W \otimes V^* \hookrightarrow \mathrm{Hom}(V, W)$$
$$\sum c_{wf} w \otimes f \mapsto \sum c_{wf} f(v)w$$

which is equivalent to say that on pure tensors our map is just

$$w \otimes f \mapsto F(w \otimes f)(v) := f(v)w$$

and for all the other elements of $V \otimes W$ we extend it linearly.

This gives us immediately that if $V$ and $W$ representations of $G$, so is $V^* \otimes W$. Since the canonical injection defined above is equivariant (prove it) we hence have that $\mathrm{Hom}(V, W)$ is a representation of $G$. What is the action of this? We have that

$$(g \circ f)(v) = \rho_W(g)\left(f\left(\rho_V(g^{-1})v\right)\right)$$

and so if $V = W$ this is just a conjugation of $f \in \mathrm{End}(V)$!

## §1.2 Subrepresentations

**Definizione 1.13** (Subrepresentation)**.** Given a representation $V$ of an object $G$, a *subrepresentation* of $V$ is a subspace $U \leq V$ such that is $G$-stable, that is

$$G \circ U \subseteq U$$

which means that acting with $G$ on $U$ always gets us to elements still in $U$.

**Esempio 1.14** (Fixed point of permutations)

Given $G = S_n$ with it's permutation representation on $V = k^n$, a subrepresentation of $V$ is

$$U = \langle e_1 + e_2 + \cdots + e_n \rangle$$

and in this case is much more than $G$ invariant: is pointwise fixed by the action of $G$. A non pointwise fixed example is the space $W$ with cartesian equation

$$\sum_{i=1}^{n} x_i = 0$$

which, if car$K \not| n$ is not pointwise fixed for every point, $w \notin G \circ w$ for any $w$.

An operation we haven't yet touched is the quotient of spaces, we want to investigate what happens if we take two representation $V \geq U$ of $G$ and consider $V/U$. We claim that this is a representation with the action

$$\rho_{V/U}(v + U) := \rho_V(g)v + U$$

and we should check that this is actually well defined and is a homomorphism. What does this mean in matrix terms? Given a basis $u_i$ of $U$ and a completion of it to a basis of $V$, the matrix of $\rho_V(g)$ in this basis has generally the form

$$\rho_V(g) = \left(\begin{array}{c|c} \rho_U(g) & * \\ \hline 0 & \rho_{V/U}(g) \end{array}\right)$$

one of the big questions of representation theory, as always when we deal with matrices, is wether, given a certain representation $V$ of a group $G$, we can express the action of $G$ on $V$ as acting through block diagonal matrices. From the formula above we see that if we can find a couple of spaces $U \oplus U'$ that are both $G$ stable, we have achieved our goal.

**Esempio 1.15** (Decomposition of a representation of $V$)

Taking $G = S_n$ and $V = k^n$, since we have

$$\langle e_1 + e_2 + \cdots + e_n \rangle \oplus \left\langle \left\{ \sum_{i=1}^{n} x_i = 0 : x \in k^n \right\} \right\rangle$$

and theese are $G$-stable we have that this is a decomposition of $V$ as a representation.